Unification of Identity and Difference: A Tri-Space Dual-Projection Framework for Transformer Interpretability

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Abstract

Research on the interpretability of large language models has long been constrained by the discreteness of neuron- and feature-centric perspectives, making it difficult to uniformly explain attention mechanisms, multi-head collaboration, and semantic emergence. Building upon our prior work on Relation Body Geometry (**?**), this paper proposes the Tri-Space Dual-Projection (TSDP) framework, which precisely aligns with the actual computational flow of Transformers (Vaswani et al., 2017). We demonstrate that: the Query, Key, and Value (Q, K, V) spaces each internally satisfy the Relation Body Unit Circle (Theorem 1); identity and difference (tong-yi) are not directional properties inherently carried by information, but rather relational properties defined by the Query-Key projection (Q-K) in the first projection; softmax normalizes the absolute intensity of QK into value shares; these shares constitute a Share Operator acting on V, allocating the proportion of each V being utilized. The boundary of identity and difference is defined by QK, while the value share is allocated by softmax; the two are decoupled but sequentially connected. The Attention-Value projection (A-V) serves as the second projection, directly acting on V due to the positional alignment between K and V. Multi-head attention consists of multiple independent TSDP modules concatenated along the feature dimension; the residual stream is the integration channel running through the layers. The architecture itself (layers, multi-head, residual) constitutes the physical skeleton of the Static Tree. Under the TSDP mechanism, semantic evolution automatically selects a path from identity to difference. Driven by semantic pressure, this evolution manifests macroscopically as a progression from a narrow-semantic Traditional Tree trunk in the shallow layers, to a broad-semantic Bush in the middle layers. In the deep layers, regardless of the macroscopic form, the demand for deep local deduction forces the attention distribution to drastically sharpen, forming a Local Petiole structure.

Combined with systematic empirical evidence covering dense/sparse architectures (Mixture of Experts, MoE), causal/bidirectional architectures (Bidirectional Encoder Representations from Transformers, BERT) (Devlin et al., 2019), pre-trained base models versus instruction-tuned models (Base/Chat) alignment paradigms, and multi-head ablation, this paper verifies that Theorem 3 (the No-Trunk Theorem) and the U-shaped contour are inherent architectural priors. **Crucially, contrasting causal interventions reveal that severing QK routing causes severe semantic collapse, while shuffling multi-head feature dimensions yields minimal impact, proving that residual streams provide a robust integration fallback.** Furthermore, within MoE architectures, the Router effectively partitions the global semantic Bush into multiple narrow-semantic Traditional Trees, while maintaining uniform orthogonality in multi-head projections. This paper provides a precise geometric framework for understanding large models, ranging from the local tangent circle to hierarchical state progression, and proposes the dual structure of the Static Tree and the Dynamic Tree. The overarching theme is the unification of identity and difference (tong-yi-gui-yi) and their evolution (tong-yi-yan-jin).

1 Introduction

1.1 The Dilemma of Large Model Interpretability and the Geometric Turn

Current research on the interpretability of large language models mostly focuses on neuron activation or feature attribution, making it difficult to uniformly explain how Transformers (Vaswani et al., 2017) process syntax, semantics, and logic in high-dimensional space. In recent years, information geometry and neural manifold theory have revealed the structural properties of representation spaces, yet there remains a lack of a unified geometric framework capable of explaining both macro-architectural priors and micro-semantic carving simultaneously. This paper posits that the operation of large models is not merely simple numerical computation, but rather a process of projection and aggregation in high-dimensional space.

1.2 Phenomenological Observations of Prior Work

In our prior work (**?**), we proposed Relation Body Geometry and established foundational theorems: Theorem 1 (Relation Body Unit Circle, $c^2 + s^2 = 1$), Theorem 2 (Redundancy-Probability Negative Correlation), Theorem 3 (No-Trunk Theorem), and Theorem 4 (U-shape and Redundancy Compensation), while observing that layer-wise semantic evolution exhibits a Tree/Bush structure. These findings provided a new perspective for understanding Transformers, but prior work remained at the level of phenomenological observation and could not answer: What is the underlying mechanism behind these phenomena? How are Query, Key, and Value (Q, K, V), multi-head attention, and residual streams unified geometrically?

1.3 Core Contributions of This Paper: The Tri-Space Dual-Projection Framework

The core contribution of this paper is the proposal of the "Tri-Space Dual-Projection (TSDP)" framework, which precisely aligns Relation Body Geometry with the actual mechanisms of Transformers. The overarching theme of this paper is the unification of identity and difference (tong-yi-gui-yi) and their evolution (tong-yi-yan-jin): from the inherent identity and difference in vector space, to the QK definition of identity and difference, to the softmax allocation of value shares, to the A-V application of shares, to the multi-head differentiation of identity and difference, to the residual stream accumulation of identity and difference, and finally to the LM Head's final definition of semantic boundaries. Every step is an expression of identity and difference.

The core points of the framework are as follows:

- The Q, K, and V spaces each internally satisfy Theorem 1, forming internal relation body circles.
- Identity and difference (tong-yi) are not directional properties inherently carried by information, but rather relational properties defined by the Query-Key projection (Q-K) in the first projection.
- The mask restricts the visible range of identity and difference, while positional encoding determines their order.
- Softmax normalizes the absolute intensity of QK into value shares.
- These shares constitute a Share Operator acting on V, allocating the proportion of each V being utilized. The boundary of identity and difference is defined by QK, while the value share is allocated by softmax; the two are decoupled but sequentially connected.
- The Attention-Value projection (A-V) serves as the second projection; due to the positional alignment between K and V, the value shares directly act on V.
- Multi-head attention consists of multiple independent TSDP modules concatenated along the feature dimension; the residual stream is the integration channel running through the layers, carrying the layer-wise evolution of spatiotemporal states.

- The architecture itself (layers, multi-head, residual) constitutes the physical skeleton of the Static Tree. Under the TSDP mechanism, semantic evolution automatically selects a path from identity to difference (first-same-then-different), emerging as the Dynamic Tree structure observed in prior work.
- From micro to macro, the unification of identity and difference is embodied in a three-tier tree structure: the Share Tree output by softmax (micro), the Dynamic Tree of hierarchical spatiotemporal progression (meso), and the Cumulative Tree of the entire architecture (macro).

1.4 Structure and Contributions of This Paper

Chapter 2 establishes the Tri-Space Dual-Projection framework. Chapters 3 through 5 validate architectural priors, cross-space identity-difference definition, multi-head orthogonality, causal intervention, generalization boundaries, alignment paradigms, functional ablation, and sparse routing through Experiments 1 to 8. Chapter 6, under the holistic view of unifying phenomena and mechanisms, inherits from prior work and proposes theoretical extensions and future directions.

2 Theoretical Foundations

2.1 Prior Foundations: Foundational Theorems of Relation Body Geometry

In our prior work (**?**), we established the following foundational theorems:

- **Theorem 1 (Relation Body Unit Circle)**: For any non-zero vectors $\mathbf{a}, \mathbf{b}$, define the normalized inner product c and orthogonal redundancy s, satisfying $c^2 + s^2 = 1$. This unit circle applies to any pair of vectors within the same space.
- **Theorem 2 (Redundancy-Probability Empirical Theorem)**: Attention probabilities exhibit a significant negative correlation with local redundancy.
- **Theorem 3 (No-Trunk Theorem)**: In high-dimensional hidden spaces, the concentration of the first principal component (PC1) approaches the random baseline $1/d_h$, indicating the absence of a global trunk.
- **Theorem 4 (U-shape and Redundancy Compensation)**: Inter-layer semantic expansion follows a U-shaped curve; pre-training counteracts the entropy-reduction trend in the middle layers through multi-head mechanisms.

Prior work also observed that layer-wise semantic evolution presents a Tree/Bush structure: highly redundant parts grow first, while low-redundancy parts are refined subsequently. These phenomena provide the starting point for this paper.

2.2 Tri-Space Dual-Projection Framework

2.2.1 Entry and Carrier: Vector Space, Embedding, and Positional Encoding

The foundational information carrier of Transformers (Vaswani et al., 2017) is the high-dimensional vector space. The Embedding layer maps discrete tokens to a set of points within this space, serving as the entry and "digesting" layer of the geometric system. Positional encoding injects positional information into each token, determining its order within the visible field. The algebraic form of positional encoding—whether Absolute Positional Encoding or Rotary Position Embedding (RoPE)—does not affect the core mechanisms of Relation Body Geometry; the crucial factor is the positional information itself.

Layer Normalization normalizes the magnitude of individual token vectors, whereas Relation Body Geometry normalizes the relationship between two tokens. These are two distinct levels of normalization. Theorem 1

($c^2 + s^2 = 1$) derives from the algebraic properties of the inner product and is independent of LayerNorm: for any two non-zero vectors, $c = \cos\theta$ and $s = |\sin\theta|$, thus $c^2 + s^2 = 1$ holds universally.

Architectural variants such as Grouped-Query Attention (GQA) and Multi-Query Attention (MQA) alter the configuration of multi-head modules but do not change the core QK mechanism. QKV are information carriers, and the addition or removal of architectural modules does not affect the core of the TSDP framework.

2.2.2 Internal Relation Body Circles of the Three Spaces

In each layer and each head of a Transformer, the input $\mathbf{X}$ undergoes three independent projections to obtain Query, Key, and Value (Q, K, V):

$$\mathbf{Q} = \mathbf{X}\mathbf{W}_Q, \quad \mathbf{K} = \mathbf{X}\mathbf{W}_K, \quad \mathbf{V} = \mathbf{X}\mathbf{W}_V$$

Q, K, and V belong to three distinct projection spaces. Within each space, the normalized inner product and normalized outer product between any two vectors satisfy Theorem 1, forming an internal relation body circle. Specifically:

- The Q space contains Q-Q relation body circles.
- The K space contains K-K relation body circles.
- The V space contains V-V relation body circles.

These three circles are independent mathematical structures. In the actual execution path of the model, only the cross-space QK computation is explicitly executed; the Q-Q and K-K internal relation body circles do not directly participate in the architectural computation.

2.2.3 First Projection: Q-K and Masking

Q and K originate from different spaces. $\mathbf{Q}\mathbf{K}^\top$ is a cross-space inner product, resulting in a scalar matrix. This matrix can be utilized to construct cross-space relation body values (c, s).

Relevance and the Macroscopic Reduction. While the geometric essence of the QK dot product is to define identity and difference (c_{ij} and s_{ij}), in the functional context of the attention mechanism, this measure directly corresponds to the **relevance** between tokens. Mathematically, the inner product is a **macroscopic reduction**: it aggregates the signed contributions of all feature dimensions into a single scalar score. This score quantifies the overall relevance, abstracting away the fine-grained feature distribution. A high score manifests as high "identity" (tong), determining which tokens should be dynamically linked.

Visibility Constraint of Masking. After the raw QK scores are computed, the mask is added to the score matrix before the softmax operation. Causal masking sets the score of the i-th token with respect to the j-th token (where $j > i$) to $-\infty$, forcing softmax to normalize only over the first i positions. Masking does not alter the calculation of QK's c and s, but it restricts the normalization scope of softmax. Masking does not determine the order (which is determined by positional encoding); rather, it defines the visibility range.

The Causal Horizon and Parallel Causal Assembly. The mask is not merely an architectural patch applied before softmax; fundamentally, it defines the **causal horizon** along the sequence (time) dimension. By setting the scores of future tokens to $-\infty$, it enforces a strict unidirectional path dependency. A profound architectural beauty of the Transformer is that it resolves the paradox between parallel computation and strict causality: although all token representations are computed simultaneously via matrix multiplications, the mask ensures that the resulting feature at position i is an **incomplete tree** that only assembles information from past stages. Because the mask prevents any future information from leaking into the current assembly, the parallel computation does not violate causality. The final semantic tree is not generated in a temporal loop, but rather parallelly assembled from these causally isolated, incomplete fragments.

Definition of Identity and Difference (tong-yi). Identity and difference are not directional properties inherently carried by information, but rather relational properties defined by QK computation within cross-space interactions. At the level of mathematical structure, any pair of vectors possesses an intrinsic measure of identity and difference (Theorem 1 applies to arbitrary vector pairs); however, in the actual execution path of the model, only QK computation is explicitly executed. The absolute symmetry of the V space ($c_{AB} = c_{BA}$) serves as the mathematical evidence for this fact.

The QK cross-space relation body possesses geometric extremes: $c = 1$ corresponds to complete alignment, $c = 0$ corresponds to complete orthogonality, and $c = -1$ corresponds to complete opposition. However, this relation body does not possess a pole with a specific token as the reference frame, because no common reference frame exists between Q and K.

When QK computation is completed (prior to masking and softmax), the Q, K, and V values are fully determined.

2.2.4 Normalization Role of Softmax

Softmax executes a normalization operation architecturally. It does not alter positional information (retained by matrix indices), nor does it change the relative order of intensity measures; it merely normalizes the absolute intensity of QK into value shares.

Temperature Factor and Capacity. The actual form of softmax includes a temperature factor $1/\sqrt{d_k}$:

$$w_{ij} = \frac{\exp(c_{ij}/\sqrt{d_k})}{\sum_k \exp(c_{ik}/\sqrt{d_k})} \tag{2.1}$$

Here, d_k is the dimension of each head, representing the **capacity** of the feature space. It is a fixed architectural hyperparameter. Mathematically, the variance of the raw QK dot product scales with d_k; therefore, $1/\sqrt{d_k}$ is the physical bandwidth determined by the architecture to stabilize the distribution. The temperature experiment in Section 5.6 demonstrates that: if the capacity is large, the feature space is spacious, allowing every key to receive a relatively rich share, manifesting a flat and composed distribution. Conversely, if the capacity is small, resources are restricted, and the model is forced to concentrate resources on extremely few keys, resulting in a sharp distribution.

Decoupling and Soft Correlation between Capacity and Semantic Nature. It is essential to distinguish spatial capacity from semantic nature. The "Bush" represents dense semantics (multiple keys are deeply evolving), while the "Traditional Tree" represents narrow semantics (only a few keys are deeply evolving). Temperature (d_k) influences how information is loaded and spatially allocated, but it does not determine what semantic content is processed. There is no strict causal relationship between spatial capacity and semantic nature, but under optimization pressure, a **soft correlation** emerges: a large capacity naturally accommodates a dense Bush; a small capacity is physically forced into sharp contraction to avoid information overload, thereby more easily manifesting the morphological characteristics of an extremely focused Traditional Tree. The experiment in Section 5.6 simultaneously confirms that when temperature changes, the V space variance remains constant (0.052789), indicating that V acts as a pure information carrier and does not change with capacity.

Architecturally, this operation forces the result of QK computation to not act directly on V; it must be transformed into value shares via softmax before weighting V.

The normalization bottleneck possesses the following properties:

1. **Boundedness and Stability.** The sum of shares is constantly 1; the aggregation of V is a convex combination.

2. **Competitiveness and Selectivity.** Different Ks compete for a fixed attention budget, automatically generating selectivity.

3. **Interpretability.** Shares are probability distributions, directly interpretable as attention allocation proportions.
4. **Gradient Properties.** The gradient form of softmax is concise, facilitating backpropagation.
5. **Separation of Position and Intensity.** Normalization applies only to intensity (values) and does not change positions (indices).

Micro-mechanism of the Normalization Bottleneck: Share Operator.

Let the directional measure between the i-th Query and the j-th Key-Value pair be c_{ij}, and the orthogonal measure be s_{ij}, satisfying $c_{ij}^2 + s_{ij}^2 = 1$. c_{ij} and s_{ij} are defined by QK computation, describing the absolute proportion of identity and difference. Softmax normalizes c_{ij} into value shares w_{ij}.

w_{ij} represents the relative share obtained by the j-th Key-Value pair under the current Query, with a sum of 1. The share expresses the connection strength between Q and this Key-Value pair.

Define the Share Operator $\mathcal{S}_{ij}$ acting on the V vector:

$$\mathcal{S}_{ij}(\mathbf{V}_j) = w_{ij}\mathbf{V}_j \tag{2.2}$$

w_{ij} is a scalar, and its effect on $\mathbf{V}_j$ is pure scaling. Scaling preserves direction while altering magnitude.

Responsibilities of the Share Operator:

- **Connection Strength.** w_{ij} represents the connection strength between Q and the Key-Value pair. A high w_{ij} indicates strong connection and greater utilization of V; a low w_{ij} indicates weak connection and lesser utilization.
- **Value Share.** w_{ij} is the normalized share. The sum of all w_{ij} is 1, and shares are mutually competitive.
- **Non-definition of Boundaries.** Identity-difference boundaries are defined by QK computation. w_{ij} merely transforms the QK-defined strength into a relative share, without redefining the boundaries.

The division of labor between identity-difference boundaries and value shares is as follows:

- **Identity-Difference Boundary.** Defined by QK computation. c_{ij} and s_{ij} describe the absolute proportion.
- **Value Share.** Allocated by softmax. w_{ij} describes the relative share of V utilized under the current Query.
- **Decoupling of the Two.** QK defines "identity/difference," while softmax defines "how much is taken."

The three tiers sequentially connect: QK outputs absolute identity/difference (c_{ij}, s_{ij}), softmax allocates relative shares (w_{ij}), and the share operator utilizes V according to these shares. The definition of identity/difference and the allocation of value shares are conceptually decoupled but computationally linked in sequence.

2.2.5 Second Projection: A-V

The positions of $\mathbf{A}$ correspond one-to-one with the positions of $\mathbf{K}$. $\mathbf{K}$ and $\mathbf{V}$ originate from different projections of the same input $\mathbf{X}$, so their positions are aligned. Therefore, $\mathbf{A}$ directly acts on the $\mathbf{V}$ at the corresponding positions, completing the second projection. Unlike the first projection (the absolute cross-space measure of Q-K), the second projection is a position-aligned weighting process. $\mathbf{V}$ is a pure information carrier; the effect of $\mathbf{A}$ on $\mathbf{V}$ is a weighted summation that does not introduce new directionality.

2.2.6 Multi-Head Independent Modules and Residual Stream Integration

In each layer, every Head possesses independent QKV and attention computations, constituting an independent TSDP module. The multi-head outputs are concatenated along the feature dimension and then projected via $\mathbf{W}_O$ back into the residual stream.

$\mathbf{W}_O$ Output Projection. The concatenated multi-head results are projected via $\mathbf{W}_O$ back to the dimension of the residual stream, completing the integration of multi-head information. From a geometric perspective, $\mathbf{W}_O$ functions as a "feature space regression" operation for multiple TSDP modules.

Residual Stream and Inter-layer Spatiotemporal Evolution. The residual stream is the integration channel running through the layers. Each Head in each layer is responsible for deduction within its own module, while the residual stream is responsible for integration and accumulation across layers. Layer-wise progression is not continuous growth; rather, each layer block acts as an independent spatiotemporal state, inheriting all information from previous states through the residual stream. The residual stream carries the inter-layer evolution of spatiotemporal states: the state $\mathbf{s}_l$ of the l-th layer is generated by the TSDP module of that layer and accumulated into the residual stream, becoming the input for the $(l+1)$-th layer.

Formally, the residual stream update for the l-th layer is:

$$\mathbf{h}_{l+1} = \mathbf{h}_l + \mathrm{TSDP}_l(\mathbf{h}_l) \tag{2.3}$$

Here, $\mathbf{h}_l$ is the residual stream state of the l-th layer, and TSDP_l is the TSDP module operation of that layer. The hierarchical state does not "grow" from the previous layer; instead, each layer serves as an independent spatiotemporal state, inheriting all information from prior states through residual accumulation.

As established in Section 2.2.3, the Transformer operates on a unified spatiotemporal fabric. While the mask serves as the horizontal causal horizon dictating sequence generation, the residual stream serves as the vertical causal horizon dictating depth evolution. The synergy of these two unidirectional mechanisms ensures that semantics are accumulated linearly without violating the fundamental causal constraints of autoregressive generation.

Spatiotemporal Isomorphism. As established in Section 2.2.3, the Transformer operates on a unified spatiotemporal fabric. While the mask serves as the horizontal causal horizon dictating sequence generation, the residual stream serves as the vertical causal horizon dictating depth evolution. The synergy of these two unidirectional mechanisms ensures that semantics are accumulated linearly without violating the fundamental causal constraints of autoregressive generation.

2.2.7 Output: LM Head and Vocabulary Mapping

The output of the final layer undergoes Final LayerNorm and is then mapped to a probability distribution over the vocabulary dimension via the LM Head (a linear layer):

$$\mathbf{p} = \mathrm{softmax}(\mathbf{W}_{\mathrm{LM}} \cdot \mathrm{LayerNorm}(\mathbf{h}_L)) \tag{2.4}$$

Here, $\mathbf{h}_L$ is the hidden state of the final layer. The LM Head maps the high-dimensional geometric representation into a probability distribution over discrete tokens, completing the final step from geometric space to semantic output. From the perspective of Relation Body Geometry, the probability of the final output represents the ultimate definition of the identity-difference boundary: up to this point, the "form" of the semantic entity is solidified by probability.

2.2.8 Architectural Emergence: Physical Skeleton of the Static Tree

The TSDP framework reveals the micro-mechanisms of Transformer operation. On top of these mechanisms, the architecture itself (layers, multi-head, residual) automatically emerges into a macroscopic Tree/Bush structure.

The Driving Force of Growth: Hierarchical Differences in Convergence Rates and Implicit Regularization Preference. Shallow-layer parameters process inputs close to raw tokens with high semantic redundancy and strong sharing; deep-layer parameters process inputs that have undergone multiple nonlinear transformations, resulting in highly differentiated semantics. Meanwhile, gradient descent implicitly prefers simple solutions with smaller parameter norms while fitting data (Zhang et al., 2017). These two mechanisms synergistically cause the model to prioritize convergence towards simple semantics with high universal commonality (shallow layers), while deep layers fit differentiated complex semantics. However, the reason simple semantics can solidify into a macroscopic "trunk" and deep semantics extend into "branches and leaves" relies on the following three architectural conditions.

The Three Architectural Conditions for Tree Morphology:

1. **Dynamic Condition (Implicit Regularization Preference):** Gradient descent's preference for simple solutions endows the network with the driving force to prioritize learning universally common information, which is the starting point for trunk formation.
2. **Accumulation Condition (Residual Connection):** Residual connections (He et al., 2016) provide a globally shared identity mapping, ensuring that the simple, shared semantics learned in the shallow layers are continuously accumulated across layers without being overwritten by subsequent layers. This is the physical skeleton that solidifies the trunk.
3. **Topological Condition (Serial Unidirectional Progression):** The layer-wise progression of Transformers is serial and unidirectional. If the progression were nonlinear (e.g., containing loops or jumps), information transmission paths would bifurcate, and the tree would become a more complex graph structure. The serial unidirectional topology ensures that information grows additively along the depth. These three conditions are indispensable, jointly making the tree morphology a natural product under the current Transformer architecture.

Incremental Addition and Training Protection Mechanism of Residual Connections. The residual stream update is formulated as $\mathbf{h}_{l+1} = \mathbf{h}_l + f_l(\mathbf{h}_l)$. It is crucial to note that the residual stream plays distinct dynamic roles across phases. During the *training phase*, it functions as an **identity mapping protection mechanism** ("do not load if it already exists"). Gradient descent implicitly suppresses the residual branch $f_l(\mathbf{h}_l)$ to zero when existing features are sufficient, preventing the overwriting of established semantic trunks (He et al., 2016). This greatly endows the network with elasticity, ensuring the static tree skeleton is solidified without degradation. Conversely, during the *inference phase*, the residual stream serves as an **incremental aggregation channel** ("do not redundantly add if it already exists"). Instead of reconstructing the entire tree at each layer, the model dynamically adds context-specific semantic increments onto the existing accumulated trunk. Empirical results showing that hierarchical representation similarity gradually increases with depth (Kornblith et al., 2019) support this interpretation. Under the TSDP framework, this is further interpreted as dynamic branch growth: shallow layers accumulate shared semantics to form the trunk, while deep layers append differentiated increments, extending into branches and leaves.

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Optimization Dynamics of "First-Same-Then-Different". Combining the above: shallow layers converge first (first-same), and deep layers converge later (then-different). The implicit regularization of gradient

descent, the accumulation of residual connections, and the serial unidirectional topology work together to make the "first-same-then-different" progression of the static tree a natural product under the TSDP mechanism.

The static tree is the physical skeleton at the architectural level, determined by the model's parametric structure and invariant to input. Its three pillars are: layers (serial structure in the depth direction), multi-head (parallel structure in the width direction), and residuals (the integration channel running through the layers).

Semantic evolution under the TSDP mechanism automatically selects a "first-same-then-different" progression path. The U-shaped contour is precisely the manifestation of this progression in inter-layer variance; it is a byproduct of architectural emergence, not a preset constraint. The above optimization dynamics argument is a theoretical derivation; its direct validation requires tracking parameter convergence rates with training checkpoints, as detailed in the validation status in Section 2.2.10.

Attention Concentration and Morphological Distribution: The experiment in Section 5.6 indicates that attention concentration increases overall with depth (entropy dropping from 1.35 to 0.68). Combined with the U-shaped contour of the V space observed in prior work, both demonstrate that: shallow layers have lower semantic pressure, presenting a clearly structured Traditional Tree morphology; middle layers experience increased semantic pressure, presenting a divergent Bush morphology. In deep layers, regardless of the macroscopic form, the demand for deep, highly specific local deduction forces the attention distribution to drastically sharpen, forming a Local Petiole structure.

Furthermore, the comparative experiments in Sections4.2.3 and 4.2.4 demonstrate that severing QK routing leads to semantic collapse, whereas shuffling feature dimensions has almost no impact. This further confirms the integration fallback capability of the residual stream, as well as the core position of QK identity-difference definition in semantic generation.

For a detailed dual structure of the Static Tree and Dynamic Tree, see Section 6.2.

2.2.9 Main Line: Unification of Identity and Difference from Vectors to Probabilities

Reviewing the complete chain of the TSDP framework, every step from vectors to probabilities is an expression of identity and difference:

- **Vector Space**: The latent carrier of identity and difference. Embedding and positional encoding map discrete tokens to high-dimensional vectors, where any two vectors possess an intrinsic relation of identity and difference (Theorem 1).
- **LayerNorm**: The geometric foundation of the relation body. Normalizing single-token vectors onto a unit sphere provides a rigorous geometric basis for pairwise relational normalization.
- **QK Computation**: The explicit definition of identity and difference. The cross-space inner product of QK determines the directional measure c_{ij} and the orthogonal measure s_{ij}.
- **Masking**: The visibility constraint of identity and difference. Causal masking restricts each Query to only see the current and preceding Keys.
- **Softmax**: The allocation of value shares. The absolute identity and difference output by QK are transformed into relative value shares. The share boundary is defined by softmax, and the identity-difference boundary is defined by QK; the two are decoupled but sequentially connected.
- **A-V Weighting**: The application of shares. Each Head independently applies value shares to V.
- **Multi-Head Concatenation**: The outputs of multiple independent TSDP modules are concatenated along the feature dimension and projected via $\mathbf{W}_O$ back into the residual stream.
- **Residual Stream**: The inter-layer accumulation of identity and difference. Each layer's output is accumulated into the residual stream through residual connections, forming the layer-wise evolution of spatiotemporal states.

- **MLP**: Nonlinear semantic transformation. Independently performs nonlinear reshaping on each token in the residual stream. MLP also appends only the newly added semantic values of the current layer via residual connections.
- **LM Head**: The ultimate definition of identity and difference. After Final LayerNorm, the final layer maps to a probability distribution over the vocabulary via a linear layer and softmax. The output probability is the quantitative expression of the semantic boundary.

Topological Scissors and Feature Filters. The operational essence of the Transformer can be understood as a two-step refinement process. First, QK and Softmax act as **topological scissors**. By assigning near-zero shares to irrelevant vectors based on a macroscopic reduction, they effectively sever the information flow from irrelevant tokens, ensuring that the subsequent aggregation is an exclusive, highly selective unification of identity. Second, the MLP acts as a **nonlinear filter**. Unlike Softmax's macroscopic share deprivation, MLP performs a **microscopic, dimension-wise screening** at the feature level. Through nonlinear activations, it independently amplifies useful feature components and prunes noisy dimensions to zero. The overarching principle is that the unification of identity and difference is achieved through macroscopic selection (Softmax) followed by microscopic feature purification (MLP).

Clarification on Residual Accumulation. The multi-head output of each layer is local, containing only the information computed by that layer's attention; it is not a re-expression of the entire information. Therefore, the accumulation of the residual stream is a layer-by-layer superposition of local increments, rather than reconstructing a complete semantic tree at each layer. This aligns with the "branch growth" interpretation in Section 2.2.8: each layer grows new branches on the existing accumulated state, rather than generating independent new trees in a nested manner.

From Micro to Macro, the unification of identity and difference is embodied in a three-tier tree structure.

- **Micro Level (Share Tree)**: The probability distribution output by softmax. High-share Keys are the trunk, low-share Keys are the leaves. Each Query corresponds to a share tree.
- **Meso Level (Dynamic Tree)**: The state sequence formed by hierarchical spatiotemporal progression. Each layer is an independent spatiotemporal state, with information inherited across layers via the residual stream.
- **Macro Level (Cumulative Tree)**: The complete tree jointly formed by the V-value matrices scattered across various layers in the entire architecture. This is the Bush/Tree structure observed in prior work.

These three tiers are nested, jointly constituting the complete geometric picture of the "unification of identity and difference." **In terms of macroscopic morphology, this evolution perfectly aligns with the optimization dynamics in Section 2.2.8.** Due to the **implicit regularization preference for simple solutions** in gradient descent, shallow layers prioritize learning foundational semantics with high commonality (the trunk of the Traditional Tree); middle layers must carry complex, multi-dimensional semantic interweaving, where semantic pressure peaks, presenting a divergent **Bush** morphology; in deep layers, to complete the extreme deduction required for final prediction, information sharply converges, and attention is highly focused, forming a **Local Petiole**.

Conceptual Clarification: Distinguishing Inter-layer Morphology from Intra-layer Semantics. When discussing the three-tier tree structure, it is crucial to strictly distinguish between "inter-layer evolutionary morphology" and "intra-layer semantic distribution." The "middle-layer Bush" mentioned earlier, in terms of **vertical layer-depth evolution**, refers to the middle layers reaching a peak in semantic expansion with the flattest attention allocation. In terms of **horizontal intra-layer distribution**, it signifies that the layer carries highly dense and dispersed semantics (extremely low PC1 concentration). Conversely, shallow and deep layers manifest as clearly structured "Traditional Trees" vertically (low semantic expansion), while deep layers contract into highly focused "Local Petioles" horizontally. The vertical progression of "first-same,

then-different, finally converging" and the horizontal distribution of "trunk-dense-focus" are two sides of the same geometric coin.

It must be emphasized that the unification of identity and difference manifests differently across mechanisms: **QK represents local unification (cross-space relational definition), whereas the residual stream represents global unification under linear progression.** The novel differences (local incremental features) generated by each layer are continuously integrated into the globally shared trunk through the linear addition of residuals. This collaboration between local directional definition and global linear accumulation jointly supports the physical skeleton of the Static Tree.

Identity and difference are inseparable. The V vector is utilized as a whole and is not partitioned into a "part of identity" and a "part of difference." "Identity" does not mean eliminating "difference"; rather, it represents the connection strength of the relationship. Strength determines the share, and the share determines how much of the entire V vector is utilized. This is the geometric essence of the unification of identity and difference.

Topological Scissors and Feature Filters. It is crucial to distinguish the operational mechanisms of Softmax and MLP. The operational essence of the Transformer can be understood as a two-step refinement process. First, QK and Softmax act as **topological scissors**. By assigning near-zero shares to highly orthogonal (irrelevant) vectors, they effectively sever the information flow from irrelevant tokens, ensuring that the subsequent aggregation is an exclusive, highly selective unification of identity. Second, the MLP acts as a **nonlinear filter**. It refines the retained semantic representations at the feature dimension level, pruning noisy dimensions through activation sparsity. The overarching principle is that the unification of identity and difference is achieved by aggressively excising the irrelevant and then finely sculpting the relevant.

2.2.10 Validation Status Statement

The validation status of the theoretical framework proposed in Section 2.2 is as follows.

Parts directly validated by existing experiments:

- Theorems 1-4 are jointly validated by prior work (**?**) and the series of empirical studies in Chapters 3 through 5 of this paper.
- The definition of identity and difference (Section 2.2.3) is indirectly supported by the QK asymmetry measurements in Section 3.2.
- The Share Operator (Equation 2.2.4) is a mathematical derivation.
- The Static Tree and Dynamic Tree (Section 2.2.8) are indirectly supported by the U-shaped contour in Section 3.1, the layer-wise intervention in Section 4.2, and the residual fallback mechanism demonstrated in Supplementary Experiment B (Section4.2.4).
- The decoupling of capacity and semantic nature (Section 2.2.4) is directly validated by the temperature experiment in Section 5.6. The experiment shows that altering temperature (capacity) only changes the attention allocation morphology, while the V-space variance remains constant, confirming their geometric orthogonality.

Parts not yet directly validated:

- The relationship between temperature and identity-difference boundaries (Section 2.2.4) exhibits a "locally sharp vs. globally flat" distributional difference in experimental results, but the universality of this difference requires confirmation on more models.
- The staged accumulation effect of the residual stream (Section 2.2.6) requires a metric capable of distinguishing semantic types, for which there is currently no standard indicator.
- The statement that the algebraic form of positional encoding does not affect the core mechanism (Section 2.2.1) is a theoretical proposition.

- The optimization dynamics argument for architectural emergence (Section 2.2.8) requires training checkpoints (e.g., Pythia (Biderman et al., 2023)) to directly validate hierarchical differences in parameter convergence rates, which is left for future work.
- The mathematical form of the LM Head (Section 2.2.7) is a standard architectural component and requires no validation; its geometric interpretation as the "ultimate definition of identity-difference boundaries" is a conceptual statement.

Validation Limitations. The common characteristic of the unvalidated parts mentioned above is that they do not alter the core mechanisms of the TSDP framework; they only affect the detailed expression of the mechanisms. The core mechanisms (identity-difference definition, Share Operator, Static Tree/Dynamic Tree) have been comprehensively validated by the series of empirical studies (including supplementary causal interventions and slicing experiments) in Chapters 3 through 5.

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A Appendix